

```
import pandas as pd

df = pd.read_csv("/content/space_missions_dataset.csv")

print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 15 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Mission ID                           500 non-null    object
 1   Mission Name                         500 non-null    object
 2   Launch Date                          500 non-null    object
 3   Target Type                          500 non-null    object
 4   Target Name                          500 non-null    object
 5   Mission Type                         500 non-null    object
 6   Distance from Earth (light-years)    500 non-null    float64
 7   Mission Duration (years)             500 non-null    float64
 8   Mission Cost (billion USD)           500 non-null    float64
 9   Scientific Yield (points)            500 non-null    float64
10   Crew Size                            500 non-null    int64
11   Mission Success (%)                  500 non-null    float64
12   Fuel Consumption (tons)              500 non-null    float64
13   Payload Weight (tons)                500 non-null    float64
14   Launch Vehicle                       500 non-null    object
dtypes: float64(7), int64(1), object(7)
memory usage: 58.7+ KB
None
```

```
print(df.describe())
```

	Distance from Earth (light-years)	Mission Duration (years)	\	
count	500.000000	500.000000		
mean	25.483060	15.736800		
std	14.942128	7.578316		
min	0.350000	1.400000		
25%	11.750000	8.900000		
50%	26.185000	16.400000		
75%	38.570000	22.200000		
max	49.900000	29.500000		
	Mission Cost (billion USD)	Scientific Yield (points)	Crew Size	\
count	500.000000	500.000000	500.000000	
mean	277.300280	55.223400	50.118000	
std	141.137422	26.446232	27.660989	
min	13.320000	10.000000	1.000000	
25%	149.960000	33.775000	27.000000	
50%	282.170000	54.400000	50.000000	
75%	399.995000	79.025000	74.000000	
max	538.320000	99.800000	99.000000	
	Mission Success (%)	Fuel Consumption (tons)	Payload Weight (tons)	\
count	500.000000	500.000000	500.000000	
mean	92.616600	2543.522140	50.355620	
std	9.391094	1492.964489	28.227546	
min	66.000000	18.060000	1.020000	
25%	85.500000	1177.315000	25.567500	
50%	98.600000	2597.985000	50.995000	
75%	100.000000	3859.355000	74.482500	
max	100.000000	5018.600000	99.780000	

```
print(df.head())
```

	Mission ID	Mission Name	Launch Date	Target Type	Target Name	Mission Type	\
0	MSN-0001	Mission-1	2025-01-01	Star	Titan	Colonization	
1	MSN-0002	Mission-2	2025-01-08	Exoplanet	Betelgeuse	Colonization	
2	MSN-0003	Mission-3	2025-01-15	Asteroid	Mars	Exploration	
3	MSN-0004	Mission-4	2025-01-22	Exoplanet	Titan	Colonization	
4	MSN-0005	Mission-5	2025-01-29	Exoplanet	Proxima b	Mining	
Distance from Earth (light-years)			Mission Duration (years)		\		
0			7.05			5.2	
1			41.76			23.0	
2			49.22			28.8	
3			26.33			17.8	
4			8.67			9.2	
Mission Cost (billion USD)			Scientific Yield (points)		Crew Size \		
0			526.68			64.3	21
1			234.08			84.4	72
2			218.68			98.6	16
3			232.89			36.0	59
4			72.14			96.5	31
Mission Success (%)			Fuel Consumption (tons)		Payload Weight (tons) \		
0			100.0			731.88	99.78
1			89.6			4197.41	45.72
2			98.6			4908.00	36.12
3			90.0			2569.05	40.67
4			73.2			892.76	12.40
Launch Vehicle							
0	SLS						
1	Starship						
2	Starship						
3	Starship						
4	Starship						

```
df.dropna(inplace=True)
df
```

	Mission ID	Mission Name	Launch Date	Target Type	Target Name	Mission Type	Distance from Earth (light-years)	Mission Duration (years)	Mission Cost (billion USD)	Scientific Yield (points)	Crew Size	Mission Success (%)	Fuel Consumption (tons)	Payload Weight (tons)	Launch Vehicle
0	MSN-0001	Mission-1	2025-01-01	Star	Titan	Colonization	7.05	5.2	526.68	64.3	21	100.0	731.88	99.78	SLS
1	MSN-0002	Mission-2	2025-01-08	Exoplanet	Betelgeuse	Colonization	41.76	23.0	234.08	84.4	72	89.6	4197.41	45.72	Starship
2	MSN-0003	Mission-3	2025-01-15	Asteroid	Mars	Exploration	49.22	28.8	218.68	98.6	16	98.6	4908.00	36.12	Starship
3	MSN-0004	Mission-4	2025-01-22	Exoplanet	Titan	Colonization	26.33	17.8	232.89	36.0	59	90.0	2569.05	40.67	Starship
4	MSN-0005	Mission-5	2025-01-29	Exoplanet	Proxima b	Mining	8.67	9.2	72.14	96.5	31	73.2	892.76	12.40	Starship
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
495	MSN-0496	Mission-496	2034-06-28	Planet	Betelgeuse	Colonization	48.26	25.3	274.82	91.2	64	96.2	4837.16	44.97	Ariane 6
496	MSN-0497	Mission-497	2034-07-05	Asteroid	Betelgeuse	Exploration	0.35	4.1	379.71	82.6	61	100.0	62.59	67.88	SLS
497	MSN-0498	Mission-498	2034-07-12	Planet	Ceres	Exploration	47.60	26.6	296.45	98.6	29	100.0	4794.01	51.38	Falcon Heavy
498	MSN-0499	Mission-499	2034-07-19	Planet	Betelgeuse	Research	31.99	18.0	457.38	77.9	39	100.0	3102.16	86.28	SLS
499	MSN-0500	Mission-500	2034-07-26	Planet	Io	Exploration	43.41	23.5	449.40	45.4	88	100.0	4302.93	86.74	Falcon Heavy
500 rows × 15 columns															

Apply standardization to the numerical columns only

df[numeric_cols]=scaler.fit_transform(df[numeric_cols]) df															
	Mission ID	Mission Name	Launch Date	Target Type	Target Name	Mission Type	Distance from Earth (light-years)	Mission Duration (years)	Mission Cost (billion USD)	Scientific Yield (points)	Crew Size	Mission Success (%)	Fuel Consumption (tons)	Payload Weight (tons)	Launch Vehicle
0	MSN-0001	Mission-1	2025-01-01	Star	Titan	Colonization	-1.234866	-1.391781	1.768698	0.343553	-1.053728	0.787000	-1.214668	1.752681	SLS
1	MSN-0002	Mission-2	2025-01-08	Exoplanet	Betelgeuse	Colonization	1.090423	0.959379	-0.306535	1.104347	0.791870	-0.321541	1.108897	-0.164388	Starship
2	MSN-0003	Mission-3	2025-01-15	Asteroid	Mars	Exploration	1.590183	1.725487	-0.415758	1.641823	-1.234669	0.637774	1.585333	-0.504822	Starship
3	MSN-0004	Mission-4	2025-01-22	Exoplanet	Titan	Colonization	0.056738	0.272523	-0.314975	-0.727614	0.321424	-0.278905	0.017116	-0.343470	Starship
4	MSN-0005	Mission-5	2025-01-29	Exoplanet	Proxima b	Mining	-1.126339	-0.863430	-1.455077	1.562337	-0.691846	-2.069625	-1.106802	-1.345977	Starship
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
495	MSN-0496	Mission-496	2034-06-28	Planet	Betelgeuse	Colonization	1.525870	1.263180	-0.017591	1.361730	0.502365	0.381956	1.537836	-0.190984	Ariane 6
496	MSN-0497	Mission-497	2034-07-05	Asteroid	Betelgeuse	Exploration	-1.683711	-1.537077	0.726330	1.036216	0.393800	0.787000	-1.663413	0.621447	SLS
497	MSN-0498	Mission-498	2034-07-12	Planet	Ceres	Exploration	1.481656	1.434894	0.135817	1.641823	-0.764222	0.787000	1.508905	0.036326	Falcon Heavy
498	MSN-0499	Mission-499	2034-07-19	Planet	Betelgeuse	Research	0.435912	0.298941	1.277195	0.858319	-0.402340	0.787000	0.374555	1.273946	SLS
499	MSN-0500	Mission-500	2034-07-26	Planet	Io	Exploration	1.200960	1.025422	1.220598	-0.371820	1.370882	0.787000	1.179646	1.290258	Falcon Heavy
500 rows × 15 columns															

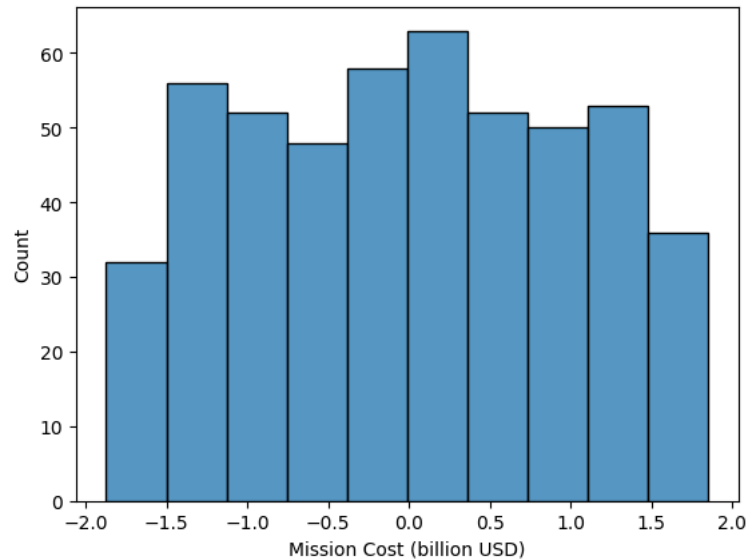
df=pd.get_dummies(df, columns=['Fuel Consumption (tons)']) df															
	Mission ID	Mission Name	Launch Date	Target Type	Target Name	Mission Type	Distance from Earth (light-years)	Mission Duration (years)	Mission Cost (billion USD)	Scientific Yield (points)	Fuel Consumption (tons)_1.5854469938515914	Fuel Consumption (tons)_1.5942235654082195	Fuel Consumption (tons)_1.595108597834097	Fuel Consumption (tons)_1.595108597834097	Fuel Consumption (tons)_1.595108597834097
0	MSN-0001	Mission-1	2025-01-01	Star	Titan	Colonization	-1.234866	-1.391781	1.768698	0.343553	...	False	False	False	False
1	MSN-0002	Mission-2	2025-01-08	Exoplanet	Betelgeuse	Colonization	1.090423	0.959379	-0.306535	1.104347	...	False	False	False	False
2	MSN-0003	Mission-3	2025-01-15	Asteroid	Mars	Exploration	1.590183	1.725487	-0.415758	1.641823	...	False	False	False	False
3	MSN-0004	Mission-4	2025-01-22	Exoplanet	Titan	Colonization	0.056738	0.272523	-0.314975	-0.727614	...	False	False	False	False
4	MSN-0005	Mission-5	2025-01-29	Exoplanet	Proxima b	Mining	-1.126339	-0.863430	-1.455077	1.562337	...	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
495	MSN-0496	Mission-496	2034-06-28	Planet	Betelgeuse	Colonization	1.525870	1.263180	-0.017591	1.361730	...	False	False	False	False
496	MSN-0497	Mission-497	2034-07-05	Asteroid	Betelgeuse	Exploration	-1.683711	-1.537077	0.726330	1.036216	...	False	False	False	False
497	MSN-0498	Mission-498	2034-07-12	Planet	Ceres	Exploration	1.481656	1.434894	0.135817	1.641823	...	False	False	False	False
498	MSN-0499	Mission-499	2034-07-19	Planet	Betelgeuse	Research	0.435912	0.298941	1.277195	0.858319	...	False	False	False	False
499	MSN-0500	Mission-500	2034-07-26	Planet	Io	Exploration	1.200960	1.025422	1.220598	-0.371820	...	False	False	False	False
500 rows × 514 columns															

df['Mission Cost (billion USD)'].mean()
np.float64(-3.552713678800501e-17)

```
import seaborn as sns
import matplotlib.pyplot as plt
```

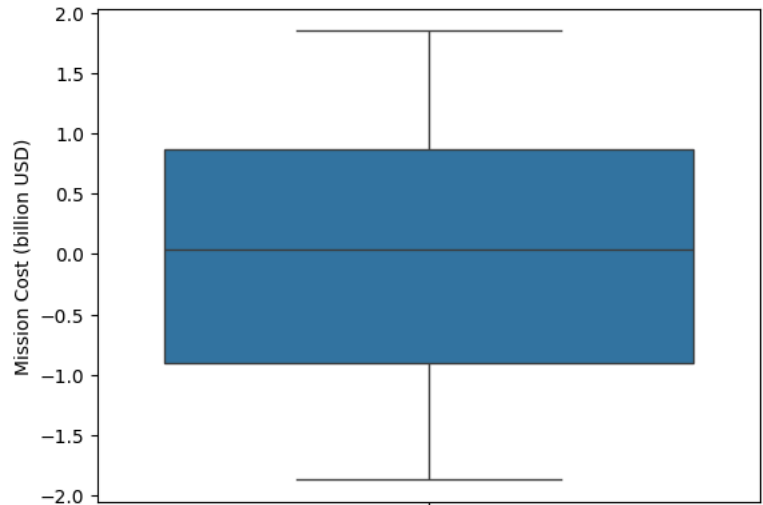
Histogram

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.histplot(df['Mission Cost (billion USD)'])
plt.show()
```



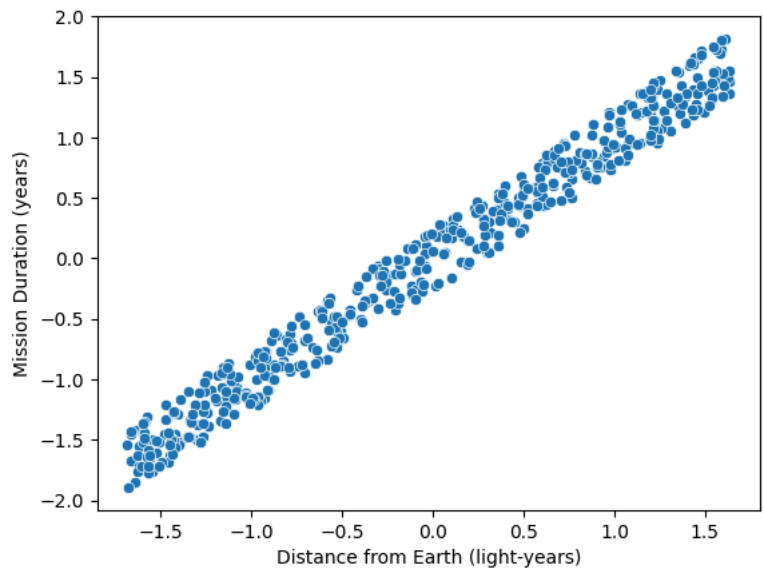
Box plot

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.boxplot(df['Mission Cost (billion USD)'])
plt.show()
```



scatter plot

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.scatterplot(x='Distance from Earth (light-years)', y='Mission Duration (years)', data=df)
plt.show()
```

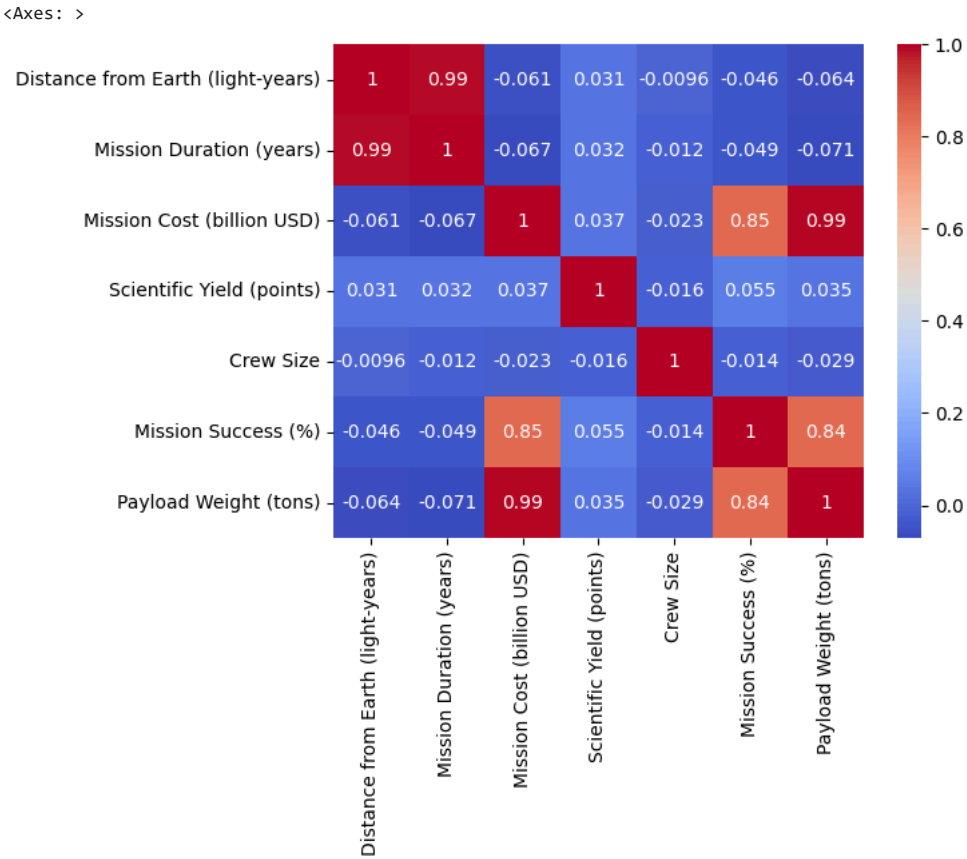


Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt

numeric_df = df.select_dtypes(include=['number'])
corr_matrix = numeric_df.corr()

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
```



```
import matplotlib.pyplot as plt
import seaborn as sns

# Ensure 'Launch Date' is a datetime object for sorting
df['Launch Date'] = pd.to_datetime(df['Launch Date'])

# Line Plot (Trend)
line_fig = plt.figure(figsize=(10, 6))
sns.lineplot(x='Launch Date', y='Mission Cost (billion USD)', data=df.sort_values(by='Launch Date'))
plt.title('Mission Cost Over Time')
plt.xlabel('Launch Date')
plt.ylabel('Mission Cost (billion USD)')
line1_figure = line_fig # Assign the figure object

# Bar Plot (Comparison) - using Mission Type counts
# Ensure mission_type_counts is defined if not already from previous execution
if 'mission_type_counts' not in locals():
    mission_type_counts = df['Mission Type'].value_counts()

bar_fig = plt.figure(figsize=(10, 6))
sns.barplot(x=mission_type_counts.index, y=mission_type_counts.values)
plt.title('Number of Missions by Type')
plt.xlabel('Mission Type')
plt.ylabel('Count')
bar1_figure = bar_fig # Assign the figure object

# Heatmap (Correlation)
# Ensure corr_matrix is defined if not already from previous execution
if 'corr_matrix' not in locals():
    numeric_df = df.select_dtypes(include=['number'])
    corr_matrix = numeric_df.corr()

heat_fig = plt.figure(figsize=(12, 10))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix of Numerical Features')
heat1_figure = heat_fig # Assign the figure object

# Pie Chart (Distribution) - using Mission Type counts
pie_fig = plt.figure(figsize=(8, 8))
plt.pie(mission_type_counts.values, labels=mission_type_counts.index, autopct='%1.1f%%', startangle=90)
plt.title('Distribution of Mission Types')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

# Close figures to prevent them from displaying inline if not explicitly requested
plt.close(line_fig)
plt.close(bar_fig)
plt.close(heat_fig)
plt.close(pie_fig)
```

```
import gradio as gr

with gr.Blocks (title="Space Mission Analytics Dashboard") as demo:
    gr.Markdown("### Space Mission Performance Dashboard")

    gr.Markdown("This dashboard provides an overview of space mission data and insights.")

    with gr.Row():
        gr.Dataframe(df, label="Source Dataset", wrap=True)

    with gr.Row():
        with gr.Column():
            gr.Plot(line1_figure, label="Mission Cost Over Time")

        with gr.Column():
            gr.Plot(bar1_figure, label="Missions by Type")

    with gr.Row():
        with gr.Column():
            gr.Plot(heat1_figure, label="Correlation Matrix")

        with gr.Column():
            gr.Plot(pie_fig, label="Distribution of Mission Types")

demo.launch()
```

It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share=True` (you can turn this off by setting `share=False` in `launch`)

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://215379e13f597441cc.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory to deploy to Hugging Face Spaces (<https://huggingface.co/spaces>)

Space Mission Performance Dashboard

This dashboard provides an overview of space mission data and insights.

Source Dataset

Mission ID	Mission Name	Launch Da...	Target Ty...	Target Na...	Mission Type	Distance from Earth (light-year...	Mission Duration (years)	Mi
MSN-0079	Mission-79	2026-07-01 00:00:00	Star	Mars	Mining	-0.8657402848173317	-0.6388812918392552	-0
MSN-0080	Mission-80	2026-07-08 00:00:00	Star	Io	Research	-0.0645172148621286	-0.24261849918341546	-1